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```
%%Section 1:
A = 1:20;
A = reshape(A,4,5)
A([1 2], [3 4 5])
A(1:3, 4:-1:2)
A(:, 3)
A(:, [3,3])
p = [2 3 5 4 1];
A(:,p)
%A(p,:) this shows error that why commented
Q=[2 3 4 1]
A(Q,:)
```

```
A =

     1     5     9    13    17
     2     6    10    14    18
     3     7    11    15    19
     4     8    12    16    20


ans =

     9    13    17
    10    14    18


ans =

    13     9     5
    14    10     6
    15    11     7


ans =

     9
    10
    11
    12
```

```
ans =

     9     9
    10    10
    11    11
    12    12


ans =

     5     9    17    13     1
     6    10    18    14     2
     7    11    19    15     3
     8    12    20    16     4


Q =

     2     3     4     1


ans =

     2     6    10    14    18
     3     7    11    15    19
     4     8    12    16    20
     1     5     9    13    17
```

Section 2:

```
B = reshape(1:9,3,3)
B*B
B.*B
B*B == B.*B
c = 5;
c*B
c+B
```

```
B =

     1     4     7
     2     5     8
     3     6     9


ans =

    30    66   102
    36    81   126
```

```
    42    96   150


ans =

     1    16    49
     4    25    64
     9    36    81


ans =

  3×3 logical array

   0   0   0
   0   0   0
   0   0   0


ans =

     5    20    35
    10    25    40
    15    30    45


ans =

     6     9    12
     7    10    13
     8    11    14
```

Section 3

```
U = ones(3,3)
Z = zeros(3,3)
5.^Z - U
R = rand(size(U))
2*R-1
R == R.'
ans - eye(3)
v = diag(R)
diag(v)


U =

     1     1     1
     1     1     1
     1     1     1
```

```
Z =

     0     0     0
     0     0     0
     0     0     0


ans =

     0     0     0
     0     0     0
     0     0     0


R =

    0.5132    0.3934    0.4768
    0.5332    0.1089    0.9413
    0.7720    0.6993    0.1527


ans =

    0.0265   -0.2131   -0.0463
    0.0663   -0.7821    0.8825
    0.5440    0.3987   -0.6945


ans =

  3×3 logical array

   1   0   0
   0   1   0
   0   0   1


ans =

     0     0     0
     0     0     0
     0     0     0


v =

    0.5132
    0.1089
    0.1527


ans =

    0.5132         0         0
```

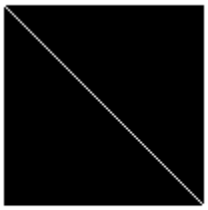
```
         0    0.1089         0
         0         0    0.1527
```

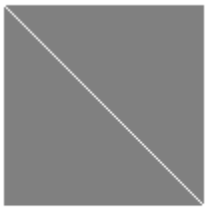
Section 4

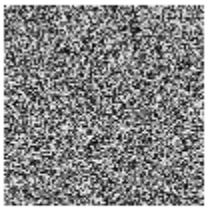
```
m = 100;
U = ones(m,m);
figure;imshow(U)
H = 0.5*U;figure; imshow(H)
Z= 0*U;figure; imshow(Z)
E = eye(m);figure; imshow(E)
figure;imshow(0.5*E + 0.5*U)
R = rand(m,m);figure; imshow(R)
figure;imshow(0.5*E + 0.5*R)
```

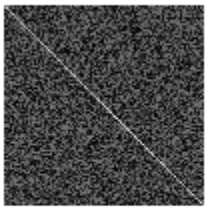










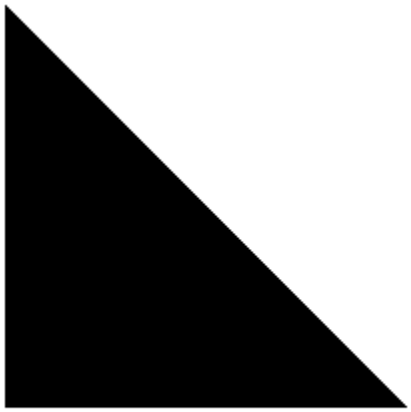


Section 5

```
m = 200;
a = 0 : 1/m : 1;
u = ones(size(a));
X = u.'*a;
Y = a.'*u;
figure;imshow(X)
figure;imshow(Y)
figure;imshow(X>Y)
figure;imshow(X.^2<Y)
C = (cos((X+Y-1)*pi/2)).^6;
figure;imshow(C)
F = a.'*a;
figure;imshow((1-F).^6)
```











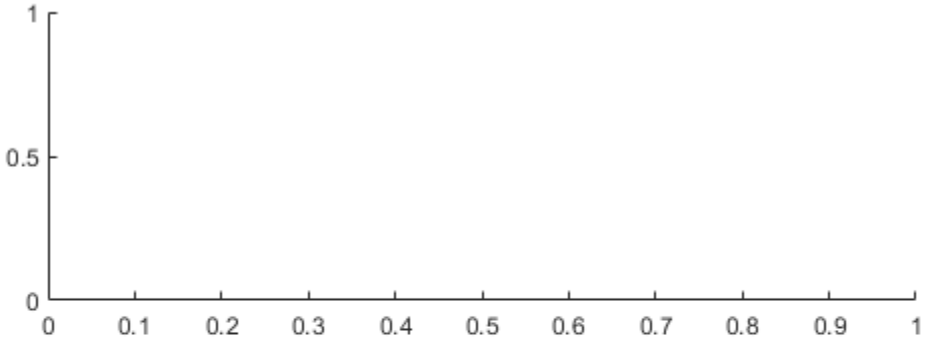


Section 6: Image transformations

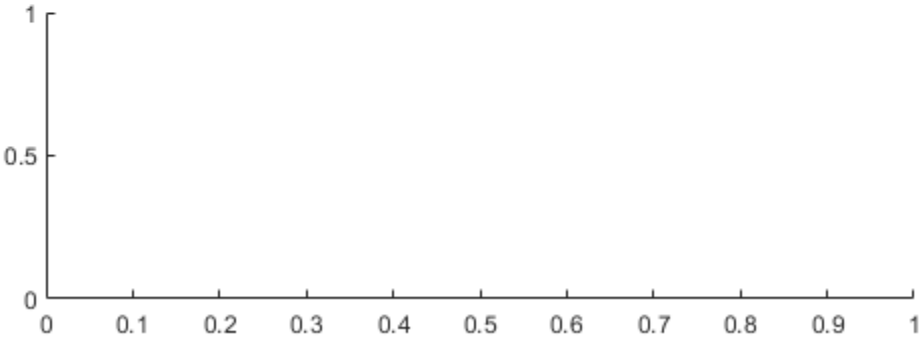
```
A = imread('cameraman.tif');
disp(size(A))
figure
figure;subplot(2,1,1),figure; imshow(A)
A1 = A(30:95, 85:150);
figure;subplot(2,1,2),figure; imshow(A1)
A2 = rot90(A,1);
figure;imshow(A2)
figure;imshow(fliplr(A))
figure;imshow(flipud(A))
figure;imshow(A.')
A3 = circshift(A,128);
figure;imshow(A3)
```

```
A4 = (circshift(A3.',128)).';
figure;imshow(A4)
A5 = A>120;
figure;imshow(A5)
A6 = double(A).^0.8;
figure;imshow(uint8(A6))
A7 = double(A).^1.05;
figure;imshow(uint8(A7))
```

512 512


1
0.5
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1




1
0.5
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1















Challenge

```
A = imread('cameraman.tif');
B = A(206:345, 377:466);% 1. zoom in (MATLAB indexes from 1)
B = flipud(B);     % 2. upside-down flip
B = double(B);% 3. convert to double
B = uint8(255*(B/255).^1.05);% 4. gamma adjustment
B = imresize(B, [135 83]);% 5. resize to the challenge
figure;imshow(B)
```



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